Sanggeon Yun

sanggeoy@uci.edu Sanggeon Yun sanggeon.com Sanggeon c0510gy
Room 3059A, Donald Bren Hall, University of California, Irvine, CA 92617

Education

University of California, Irvine *Jul. 2023 – Present*
Ph.D. Candidate in Computer Science

- GPA: **3.99/4.0**
- ICS Dean's Recruitment Fellowship (Merit-based entrance award granted upon admission)

University of California, Irvine *Jul. 2023 – Jun. 2025*
Master of Science in Computer Science

Kookmin University *Mar. 2019 – Feb. 2023*
Bachelor of Science in Computer Science

- GPA: **4.49/4.5**, Major GPA: **4.5/4.5**
- Graduate with the highest honor (**1st/145** in College of Computer Science)
- Four-year, full-ride merit scholarship (Special Talent Scholar)
- Selected to receive the President's Award for Outstanding Contribution at graduation

Research Experience

Graduate Student Researcher *Irvine, CA*
Bio-Inspired Architecture and Systems (BIASLab), University of California, Irvine (Advisor: Dr. Mohsen Imani) *Jul. 2023 – Present*

- Mainly study on overcoming challenges related to transparency, commonsense reasoning, hardware noise susceptibility, and power consumption inherent in neural-network-based models by introducing HDC (Hyperdimensional Computing)

Undergraduate Research Intern *Remote*
Bio-Inspired Architecture and Systems (BIASLab), University of California, Irvine (Advisor: Dr. Mohsen Imani) *Jan. 2022 – Jul. 2023*

- Study Hyperdimensional Computing (HDC) based brain-inspired learning models such as NeuralHD, and QuantHD to make robust models for low-power edge devices

Undergraduate Research Intern *Seoul, South Korea*
Human-Computer Interaction Laboratory (HCIL), Seoul National University (Advisor: Dr. Jinwook Seo) *Jun. 2022 – Oct. 2022*

- Scalable large-scale graph visualization and evaluation algorithms on the distributed system

Undergraduate Research Intern *Seoul, South Korea*
Natural Language Processing Laboratory, Kookmin University (Advisor: Dr. Seung-Shik Kang) *Dec. 2021 – Jun. 2022*

- Design and Development of Korean hate speech and bias detection model using transformer models
- Comparative study of various classification models for natural language by using sentiment analysis of Korean movie reviews

Undergraduate Research Intern *Seoul, South Korea*
Information Security Lab, Kookmin University (Advisor: Dr. MyungKeun Yoon) *2019 – Jun. 2020*

- Studied probabilistic data structures such as LSH (Locality Sensitive Hashing), Simhash, Bloom filter, Aging Bloom Filter (such as two active buffer bloom filter, and Segmented Aging Bloom Filter), and Cuckoo filter.

Work Experience

Software Engineer — *Seoul, South Korea*
RiderDash — *Jun. 2020 – Aug. 2021*

- **Web development**: Used TypeScript, React.js, MobX, Redux.js, Apollo + GraphQL
- **Research projects**
  - Improvement of the image recognition system for car number & VIN (Vehicle Identification Number) from a given document or car image—from architecture design to publishing (using AWS) & hypothesis testing (with 95% conf)
  - Development of the task-oriented chatbot system for sales using BERT-based neural reading comprehension model

Full Stack Engineer — *San Francisco, CA*
51 Foundation — *Dec. 2019 – May 2021*

- Made social web app using React.js, and Apollo + GraphQL

Publications

***equal contribution, C=Conference, J=Journal, A=Awarded paper, S=In Submission**

[**S.1**] **S. Yun**, R. Hassan, R. Masukawa, S. Jeong, M. Imani, "Practical Test-Time Adaptation for Large Language Models via Attention-Head Alignment," *preprint*

[**S.2**] **S. Yun**, R. Hassan, R. Masukawa, S. Jeong, M. Imani, "HopFormer: Sparse Graph Transformers with Explicit Receptive Field Control," *arXiv preprint*

[**S.3**] **S. Yun**, R. Masukawa, S. Jeong, W. Huang, H. Chen, M. Imani, "Fair Context Learning for Evidence-Balanced Test-Time Adaptation in Vision-Language Models," *arXiv preprint*

[**S.4**] **S. Yun**, R. Hassan, R. Masukawa, N. Bastian, M. Imani, "MissionHD: Hyperdimensional Refinement of Distribution-Deficient Reasoning Graphs for Video Anomaly Detection," *arXiv preprint*

[**C.1**] **S. Yun**, R. Masukawa, S. Jeong, H. Oh, N. Bastian, M. Imani, "Learning Recoverable Neural Networks against Weight Corruption via Simple Zero-Sum Projection," *The Fortieth Annual Conference on Neural Information Processing Systems, 2026 (**NeurIPS 2026, Acceptance rate=25.7%**)*

[**J.1**] **S. Yun**, R. Masukawa, H. Oh, N. Bastian, M. Imani, "A Few Large Shifts: Layer-Inconsistency Based Minimal Overhead Adversarial Example Detection," *Transactions on Machine Learning Research, 2026 (**TMLR 2026**)*

[**C.2**] **S. Yun**, H. Oh, R. Masukawa, R. Hassan, M. Imani, "Qubit-Efficient Quantum Search for Hyperdimensional Decomposition via Logarithmic Encoding," *IEEE/ACM International Conference on Computer-Aided Design, 2026 (**ICCAD 2026, Acceptance rate=23.78%**)*

[**C.3**] **S. Yun**, R. Masukawa, M. Na, H. Oh, Y. Yamaguchi, W. Huang, S. Jeong, M. Imani, "FusionSense: Tri-Stage Near-Sensor Learning for Runtime-Adaptive Multimodal Edge Intelligence," *IEEE/ACM International Symposium on Low Power Electronics and Design, 2026 (**ISLPED 2026—Oral, Acceptance rate=36%, Oral acceptance rate=24.4%**)*

[**C.4**] **S. Yun**, R. Masukawa, S. Jeong, M. Imani, M. Imani, "NeuroHash: A Hyperdimensional Neuro-Symbolic Framework for Spatially-Aware Image Hashing and Retrieval," *IEEE International Conference on Acoustics, Speech, and Signal Processing, 2026 (**ICASSP 2026—Oral, Acceptance rate=40%, Oral acceptance rate=7.08%**)*

[**C.5**] W. Huang, S. Fu, Y. Jin, Y. Ni, Z. Cui, H. Chen, Y. He, Y. Liu, **S. Yun**, S. Jeong, R. Masukawa, W.Y. Chung, M. Imani, "MERIT: Multi-domain Efficient RAW Image Translation," *IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 (**CVPR 2026, Acceptance rate=25.42%**)*

[**C.6**] **S. Yun**, H. Oh, R. Masukawa, P. Mercati, N. Bastian, M. Imani, "LogHD: Robust Compression of Hyperdimensional Classifiers via Logarithmic Class-Axis Reduction," *IEEE/ACM Design Automation and Test in Europe Conference, 2026 (**DATE 2026, Acceptance rate=24.8%**)*

[**C.7**] **S. Yun**, H. Oh, R. Masukawa, M. Imani, "DecoHD: Decomposed Hyperdimensional Classification under Extreme Memory Budgets," *IEEE/ACM Design Automation and Test in Europe Conference, 2026 (**DATE 2026, Acceptance rate=24.8%**)*

[**C.8**] H. Oh, H. Chen, **S. Yun**, Y. Ni, W. Huang, T. Das, S. Jang, M. Imani, "QUILL: An Algorithm-Architecture Co-Design for Cache-Local Deformable Attention," *IEEE/ACM Design Automation and Test in Europe Conference, 2026 (**DATE 2026, Acceptance rate=24.8%**)*

[**C.9**] H. Oh, K. Nam, R. Bhattacharjya, H. Chen, T. Das, **S. Yun**, S. Jang, A. Ding, N. Dutt, M. Imani, "T-SAR: A Full-Stack Co-design for CPU-Only Ternary LLM Inference via In-Place SIMD ALU Reorganization," *IEEE/ACM Design Automation and Test in Europe Conference, 2026 (**DATE 2026, Acceptance rate=24.8%**)*

[**C.10**] S. Jeong, H. Chen, **S. Yun**, S. Cho, W. Huang, X. Liu, M. Imani, "Cross-Modal Event Encoder: Bridging Image–Text Knowledge to Event Streams," *IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 (**WACV 2026, Acceptance rate=33%**)*

[**J.2**] **S. Yun**, R. Masukawa, R. Hassan, M. Na, M. Imani, "Contextual Fusion Strategies for Multimodal GNN-based Reasoning: Performance and Computational Trade-offs," *IEEE Access, 2026*

[**C.11**] **S. Yun**, R. Masukawa, M. Na, M. Imani, "MissionGNN: Hierarchical Multimodal GNN-based Weakly Supervised Video Anomaly Recognition with Mission-Specific Knowledge Graph Generation," *IEEE/CVF Winter Conference on Applications of Computer Vision, 2025 (**WACV 2025, Acceptance rate=37.8%**)*

[**C.12**] R. Masukawa*, **S. Yun***, M. Na, M. Imani, "PV-VTT: A Privacy-Centric Dataset for Mission-Specific Anomaly Detection and Natural Language Interpretation," *IEEE/CVF Winter Conference on Applications of Computer Vision, 2025 (**WACV 2025, Acceptance rate=37.8%**) (*equal contribution)*

[**J.3**] R. Masukawa, **S. Yun**, S. Jeong, N. Bastian, M. Imani, "TriageHD: A Hyper-Dimensional Learning-to-Rank Framework for Dynamic Micro-Segmentation in Zero-Trust Network Security," *IEEE Access, 2025*

[**C.13**] H. Chen, W. Huang, Y. Ni, **S. Yun**, Y. Liu, F. Wen, A. Velasquez, H. Latapie, M. Imani, "TaskCLIP: Extend Large Vision-Language Model for Task Oriented Object Detection," *European Conference on Computer Vision Workshop, 2025 (**ECCV 2025 Workshop**)*

[**J.4**] R. Masukawa, **S. Yun**, S. Jeong, W. Huang, Y. Ni, I. Bryant, N. Bastian, M. Imani, "PacketCLIP: Multi-Modal Embedding of Network Traffic and Language for Cybersecurity Reasoning," *Frontiers in Artificial Intelligence, 2025*

[**C.14**] **S. Yun**, R. Masukawa, W.Y. Chung, M. Na, N. Bastian, M. Imani, "Continuous GNN-based Anomaly Detection on Edge using Efficient Adaptive Knowledge Graph Learning," *IEEE/ACM Design Automation and Test in Europe Conference, 2025 (**DATE 2025, Acceptance rate=25%**)*

[**C.15**] S. Jeong, H.E. Barkam, **S. Yun**, Y. Kim, S. Angizi, M. Imani, "Exploiting Boosting in Hyperdimensional Computing for Enhanced Reliability in Healthcare," *IEEE/ACM Design Automation and Test in Europe Conference, 2025 (**DATE 2025 Acceptance rate=25%**)*

[**J.5**] **S. Yun**, R. Masukawa, H. Chen, S. Jeong, W. Huang, A. Rezvani, M. Na, Y. Yamaguchi, M. Imani, "Hyperdimensional Intelligent Sensing for Efficient Real-Time Audio Processing on Extreme Edge," *IEEE Access, 2025*

[**J.6**] **S. Yun***, H. Chen*, R. Masukawa, H. E. Barkam, A. Ding, W. Huang, A. Rezvani, S. Angizi, M. Imani, "HyperSense: Hyperdimensional Intelligent Sensing for Energy-Efficient Sparse Data Processing," *Advanced Intelligent Systems. 2400228 (**Wiley-VCH, IF 7.4**), 2024 (*equal contribution)*

[**S.5**] **S. Yun**, "Scalable Readability Evaluation for Graph Layouts: 2D Geometric Distributed Algorithms," *arXiv preprint, 2024*

[**C.16**] H. E. Barkam*, **S. Yun***, P. R. Genssler, H. Chen, A. Mema, A. Ding, H. Amrouch, and M. Imani, "Reliable Hyperdimensional Reasoning on Unreliable Emerging Technologies," *IEEE/ACM International Conference on Computer-Aided Design, 2023 (**ICCAD 2023, Acceptance rate=22.93%**) (*equal contribution)*

[**C.17**] S. Shou, C. Liu, **S. Yun**, Z. Wan, K. Ni, M. Imani, X. Sharon Hu, J. Yang, C. Zhuo, and X. Yin, "SEE-MCAM: Scalable Multi-bit FeFET Content Addressable Memories for Energy Efficient Associative Search," *IEEE/ACM International Conference on Computer-Aided Design, 2023 (**ICCAD 2023, Acceptance rate=22.93%**)*

[**C.18**] **S. Yun**, H. E. Barkam, H. Latapie, P. R. Genssler, H. Amrouch, and M. Imani, "Hyperdimensional Computing for Robust and Efficient Unsupervised Learning," *IEEE/ACM Asilomar Conference on Signals, Systems, and Computers, 2023 (**ACSSC 2023**)*

[**C.19**] H. E. Barkam, **S. Yun**, P. R. Genssler, Z. Zou, C. Liu, H. Amrouch, and M. Imani, "HDGIM: Hyperdimensional Genome Sequence Matching on Unreliable Highly-Scaled FeFET," *IEEE/ACM Design Automation and Test in Europe Conference, 2023 (**DATE 2023, Acceptance rate=24.6%**)*

[**J.7**] **S. Yun**, S. Kang, and H. Kim, "BERT-based Logits Ensemble Model for Gender Bias and Hate Speech Detection," *Journal of Information Processing Systems, 2023 (**JIPS 2023**)*

[**C.20**] **S. Yun**, S. Kang, and H. Kim, "BERT-based Pretraining Model for Gender Bias and Hate Speech Detection," *The 16th International Conference on Multimedia and Ubiquitous Engineering, 2022*

[**C.21**] **S. Yun**, and S. Kang, "A Comparative Study on the Sentiment Analysis of Korean Movie Reviews by Using Various Embedding Models," *The 2022 Spring Conference of Korea Multimedia Society, 2022*

[**A.1**] **S. Yun**, "Development of encryption algorithm that encrypts data through 2nd or higher dimensional torus," *The 24th Samsung Humantech Paper Award, 2017*

Professional Service

Reviewer

— Thirtieth Annual Conference on Artificial Intelligence and Statistics (AISTATS 2027) *2026*
— The Fifteenth International Conference on Learning Representations (ICLR 2027) *2026*
— The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026) *2026*
— European Conference on Computer Vision (ECCV 2026) *2026*
— International Conference on Machine Learning (ICML 2026) *2026*
— Applied Computing and Informatics *2026*
— IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026) *2026*
— ACM Transactions on AI Security and Privacy *2025*
— ACM SIGKDD Conf. on Knowledge Discovery and Data Mining (KDD 2025, Outstanding Reviewer) *2024*

Session Chair

— Asilomar Conf. on Signals, Systems, and Computers (ACSSC 2023) *Oct. 30, 2023*

Honors & Awards

KDD25 Outstanding Reviewer *2024*

- Recognition for the top 10% of reviewers

UC Irvine ICS Dean's Recruitment Fellowship *2023*

- The Graduate Dean's Recruitment Fellowship program is designed to supplement financial support packages of prospective doctoral and M.F.A. students who have received competitive offers from equivalent institutions
- One of the most prestigious fellowships at UCI ($20K)

NAVER CLOVA AI RUSH 2022 Round 2 (Final round) Qualifier *2022*

The 2022 KMMS Excellent Paper Presentation Award *2022*

- The 2022 Spring Conference of Korea Multimedia Society

The 5th Samsung Collegiate Programming Cup (SCPC) Finalist *2019*

- Over 5 thousand people participated worldwide, and only 138 (Top ∼2%) people advanced to the final through 2 rounds.

The 4th KMU Doorithon (Hackathon) 2nd prize *2019*

- A total of 10 teams participated
- Developed an android app called "Eye Keeper" that helps to protect the eyes of children who use phones.

ACM-ICPC Asia Seoul Regional Contest *2019 – 2020*

- 2020: 15th place
  - 365 teams participated nationwide, and only 60 (Top ∼16%) teams advanced to Regional Contest.
  - 1st place among 11 teams from Kookmin Univ. in the Preliminary Contest
- 2019: 18th place
  - 698 teams participated nationwide, and only 86 (Top ∼12%) teams advanced to Regional Contest.
  - 1st place among 60 teams from Kookmin Univ. in the Preliminary Contest.

Google Code Jam Round 2 Qualifier *2019 – 2020*

- 2020 Round 2: 1,980th (Top ∼2.0% out of all registrants)
  - Google Code Jam Round 2 Qualified on Round 1A
- 2019 Round 2: 2,443rd (Top ∼3.3% out of all registrants)
  - Google Code Jam Round 2 Qualified on Round 1A

Codeforces best finishes (Handle: c0510gy) *2019 –*

- Candidate Master on Codeforces (Max rating 1925—Top 5.78% on Codeforces worldwide as of July 2020)
- Best Finishes
  - Div3 Rank: 133rd / 3,743 (Top ∼3.5%)
  - Div2 Rank: 187th / 11,455 (Top ∼1.6%)

Full-tuition Scholarship *2019 – 2022*

- 4 years of full-tuition scholarships from Kookmin University (₩34.5 Million)
- Merit-based undergraduate scholarships for special talent in computer science

The 24th Samsung Humantech Paper Award Bronze prize (2nd) *2017*

- Selected among over 100 papers (Top ∼2%)
- Awarded ₩3 Million (about $2.7K)
- Sanggeon Yun, "Development of encryption algorithm that encrypts data through 2nd or higher dimensional torus", 2017 (Single-authored paper)

The 34th Korea Olympiad in Informatics National Contest Bronze prize *2017*

Personal Projects

Capstone Project: PaperFlow *Jan. 10, 2021 – May 22, 2021*

- https://kookmin-sw.github.io/capstone-2021-8
- The ***"PaperFlow"*** is a service that helps researchers catch up on current research flow by the algorithm that extracts references from papers and estimates the similarity between those papers.
- Paperflow graphs are generated by making weighted directed acyclic graphs (weighted DAG) where vertices represent papers, edges represent citations, and weights are calculated by doc2vec embedding models, and similarity functions (such as cosine similarity). Once weighted DAG is generated, the algorithm serializes papers by topological sorting after pruning weak edges.
- Tools Used: Javascript, React, Python, Django, NLP

Hackathon: Eye keeper *May 24, 2019 – May 25, 2019*

- According to several studies, using smartphones can harm kids' eye health. To prevent this, kids must keep a distance from the smartphone screen and blink frequently while using the smartphone.
- ***"Eye Keeper"*** helps this by using face recognition with the ML kit and blink detection using OpenCV.
- Using face recognition, the app estimates the distance from the screen to a kid's eyes and determines whether the kid is maintaining a proper distance from the screen or not. If not, the app shows a pop-up message.
- Tools Used: Android Java, OpenCV-Python

Home PT *Jun. 2019*

- This project aimed at developing a web service that enables users to experience PT in their homes. To achieve this, the service had to provide training functions such as pose instructions.
- To make the function, I built exercise pose recognition using pose estimation.
- I used TensorFlow.js to get points of body joints. And I built an MLP (Multi-layered Perceptron) model that takes points of body joints as input, puts pose classification results as output.
- Tools Used: TensorFlow.js, MLP, Javascript, HTML, CSS

Langtopia *Dec. 2019*

- This project is aimed at developing software that helps you to understand any language on your computer screen or audio sounds from your computer. To do this, I made two major functions: Screen Translator and Audio Subtitles.
- Screen Translator is the function that recognizes text from a computer screen using Pytesseract and then translates those recognized texts using Google Translate API and shows translated results to a user.
- Subtitles Recognizer recognizes subtitles in real-time by recognizing a certain length of sound using k-d tree data structure and Euclidian distance between the feature vector and recorded sound vector.
- Tools Used: Python, PyQt, Text Recognition, Audio Recognition

Teaching Experience

Teaching Assistant

- *"ICS 31: Introduction to Programming"* — Spring 2025, UC Irvine
- *"CS 147: Internet-of-Things. Software and Systems"* — Fall 2023, UC Irvine
- *"Scientific and Computational Thinking"* — 2019, Kookmin University
- *"Introduction to Creative Engineering Design"* — 2020 – 2021, Kookmin University
- *"File Processing"* — 2021 – 2022, Kookmin University
- *"Application of Database and XML"* — 2022, Kookmin University

Private Algorithm Tutoring

- One-on-one coaching for competitive-programming contests and technical-interview coding tests

Leadership

Student Club Leader of KPSC (Kookmin univ Problem Solving Club) *2021 – 2022*

Tech. Skills

Programming: C/C++, Python, TypeScript, JavaScript, Dart, C#, Java, MySQL, Processing

Machine Learning: PyTorch, TensorFlow

Computer Vision: OpenCV

Web Frameworks & Tools: React.js, Flutter, MobX, Node.js, Apollo GraphQL, Django, Material UI, Bootstrap, PHP